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Course: Cybersecurity

Date: March 2, 2026

Networking Security Analysis – Malware Spread in a Multi-Branch Bank

Introduction

In modern organizations, especially financial institutions, network design plays a critical role in ensuring security and operational continuity. In this scenario, a microfinance bank with multiple branches across Lagos experienced a widespread malware attack due to poor networking decisions.

1. Networking Concepts That Contributed to the Disaster

1.1 Lack of Network Segmentation

All branches were connected without proper isolation, allowing unrestricted communication. This created a flat network, making it easy for malware to spread across all systems.

1.2 Excessive Trust Between Branches

The network allowed unrestricted access between branches. This enabled the malware to move freely and access shared resources across the organization.

1.3 Absence of Internal Security Controls

There were no strong controls such as access restrictions or monitoring, allowing the malware to scan and infect multiple systems.

2. Networking Practice That Would Have Prevented the Spread

The use of network segmentation (such as VLANs or subnetting) would have isolated each branch. This would have ensured that the malware remained in the Ikeja branch and did not spread to others.

3. Type of Network Used

The bank used a Wide Area Network (WAN) to connect all branches. While this enabled fast communication, it also allowed malware to spread rapidly

4. Role of Firewall and Network Monitoring

Firewall

A properly configured firewall should have restricted communication between branches and blocked suspicious traffic.

Network Monitoring System

A monitoring system should have detected unusual activity early, alerted administrators, and allowed quick isolation of the infected branch.

5. Malware Remediation Strategy

In addition to preventing such attacks, it is important for organizations to have a clear remediation plan in case a security breach occurs. The following steps outline how the malware incident could have been effectively managed:

5.1 Immediate Containment

The first step is to isolate the infected system and branch from the rest of the network. Disconnecting the Ikeja branch would have prevented the malware from spreading to other branches.

5.2 Malware Identification and Analysis

Security tools such as antivirus software should be used to identify the type of malware. Understanding its behavior helps in determining the most effective removal method.

5.3 Malware Removal

The infected systems should be cleaned using trusted security tools. In severe cases, systems should be completely wiped and the operating system reinstalled to ensure total removal.

5.4 Network-Wide Scanning

All systems across the bank's network should be scanned to detect any hidden infections. This ensures that no part of the network remains compromised.

5.5 System Restoration

Data and services should be restored using clean and secure backups. Care must be taken to avoid restoring infected files.

5.6 Strengthening Security Controls

After remediation, stronger security measures should be implemented. These include network segmentation, stricter firewall rules, and improved access control policies.

5.7 User Awareness and Training

Since the attack originated from a phishing email, staff training is essential. Employees should be educated on how to recognize and avoid suspicious emails and links.

Summary

A proper remediation strategy focuses on containment, removal, recovery, and prevention. By following these steps, the bank would have minimized damage and restored operations more efficiently.

Conclusion

The incident was caused by poor network design and lack of security controls. Implementing segmentation, firewalls, and monitoring systems would have significantly reduced the impact of the attack.